SEQUENCE 1

Arithmetic

2–11 September 2026

1.1 Integers, multiples and divisors

Key point · Definitions. $\mathbb{N} = \{0, 1, 2, \ldots\}$ is the set of natural numbers. $\mathbb{Z}$ contains the natural numbers and their opposites. For $a, b \in \mathbb{Z}$, saying that a is a **multiple of** b means that there exists $k \in \mathbb{Z}$ such that $a = bk$. We also say that b divides a.

$18 = 6 \times 3$: 18 is a multiple of 3 and of 6. Every integer is a multiple of 1 and of itself; 0 is a multiple of every integer. The only multiple of 0 is 0.

Example 1. List the positive divisors of 18 and of 24. Write three multiples of 7, one of them negative. Is 35 a multiple of 5? of 6? Justify.

Key point · Sum of multiples. The sum of two multiples of an integer a is a multiple of a.

Proof to write.

Fluency · 5–10 min. Calculate $\frac{2}{3} + \frac{1}{6}$; $3(x+2) - x$; the image of 2 under $x \mapsto 2x - 3$.

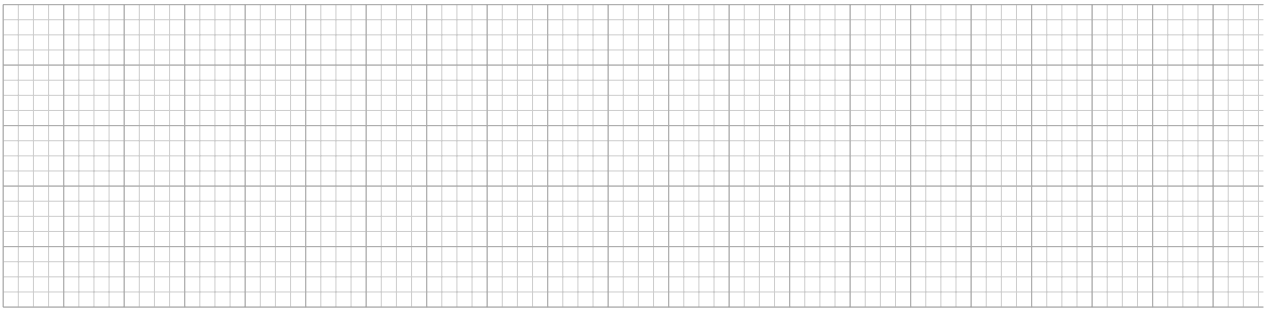
1.2 Parity and prime numbers

Key point · Definitions. An integer n is **even** if it can be written $2k$, with $k \in \mathbb{Z}$, and **odd** if it can be written $2k + 1$. A **prime number** is a natural number greater than or equal to 2 that has exactly two positive divisors: 1 and itself.

The prime numbers less than 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

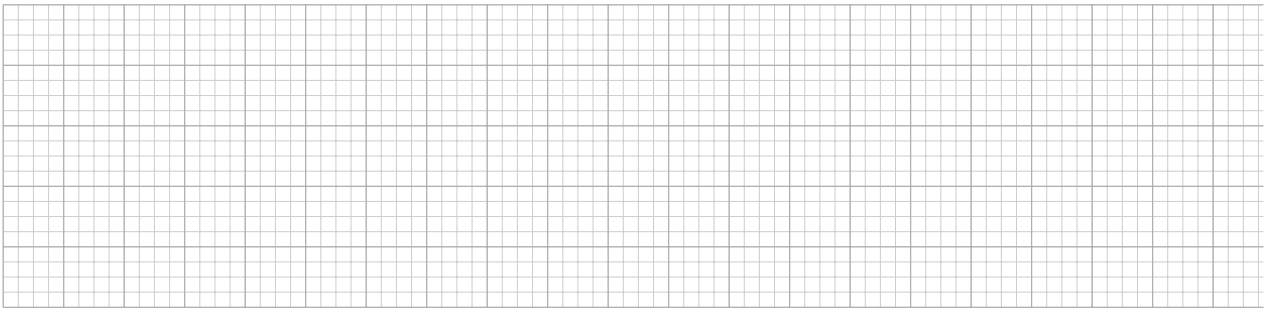
Key point · Property to prove. The square of an odd integer is odd.

Proof to write.



Method · Testing whether an integer is prime. For $n \geq 2$, it is enough to test the prime divisors less than or equal to $\sqrt{n}$. Indeed, if $n = ab$ with $1 < a \leq b$, then $a \leq \sqrt{n}$; a has a prime divisor which also divides n.

Example 2. Show that 157 is prime and that 161 is not.



1.3 GCD and irreducible fractions

Key point · Definition and convention. For two integers a, b **not both zero**, $\gcd(a, b)$ is the greatest of their common positive divisors. We have $\gcd(a, b) = \gcd(|a|, |b|)$ and $\gcd(a, 0) = |a|$ if $a \neq 0$. By convention, we set $\gcd(0, 0) = 0$; this last case does not come under the definition by the greatest divisor.

For example, $\gcd(18, 24) = 6$, and changing the signs of 18 or 24 does not change this result.

Key point · Prime factorisation – accepted. Every natural number greater than or equal to 2 is a product of prime numbers. This factorisation is unique up to the order of the factors.

Example 3. Factorise 140, 150, 630 and 120 into prime factors.

Method · Making a fraction irreducible. For $b \neq 0$, divide the numerator a and the denominator b by $\gcd(a, b)$. The fraction obtained has a numerator and a denominator that are coprime.

Example 4. Make $\frac{140}{630}$, $\frac{84}{126}$ and $\frac{-120}{150}$ irreducible.

1.4 Set vocabulary in context

Key point · Definitions and notation. $x \in A$ means that x is an element of A; $A \subseteq B$ means that every element of A belongs to B. $\varnothing$ is the empty set.

- $A \cap B$: elements belonging to A **and** to B;
- $A \cup B$: elements belonging to A **or** to B (possibly both);
- in a reference set E, $\overline{A} = E \setminus A$ contains the elements of E that are not in A;
- for a finite set, $\text{Card}(A)$ is its number of elements.

Example 5. In $E = \{1, 2, 3, 4, 5, 6\}$, let $A = \{2, 4, 6\}$ and $B = \{3, 6\}$. Give $A \cap B$, $A \cup B$, $\overline{A}$ and $\text{Card}(A)$. Are the statements $2 \in A$, $\{2\} \subseteq A$ and $A \subseteq B$ true?

Key point · Pair and Cartesian product. A pair (a, b) is ordered: in general $(a, b) \neq (b, a)$. The Cartesian product $A \times B$ is the set of pairs (a, b) with $a \in A$ and $b \in B$. For finite sets, $\text{Card}(A \times B) = \text{Card}(A)\,\text{Card}(B)$.

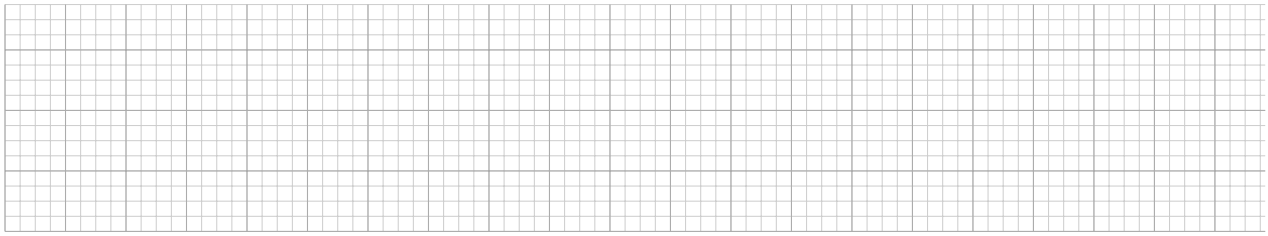
Example 6. Write all the elements of $\{1, 2\} \times \{3, 4, 5\}$.

1.5 Algorithms and putting it to use

Method · Python: integer, Boolean and condition. `a % b` is the remainder of the Euclidean division of `a` by `b` for every integer `a` when $b > 0$, the case used here (if $b < 0$, Python returns a result with the sign of b, which is not the Euclidean remainder). A test returns a Boolean: `True` or `False`. Equality is tested with `==`; assignment uses `=`.

```
def est_multiple(a, b):
    if b == 0:
        return a == 0
    return a % b == 0
```

Example 7. Predict the results of `est_multiple(35, 5)` and `est_multiple(35, 6)`, then of `est_multiple(0, 7)`, `est_multiple(7, 0)` and `est_multiple(0, 0)`. Explain the role of the condition.



Example 8 · Problem. We have 84 notebooks and 126 pencils. We want to make as many identical packs as possible, using all the material. How many packs can we make? What does each contain?

